Flat Knitting

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1. Introduction

1.1 Purpose of the study

This document describes the life cycle inventory (LCI) model developed for flat knitting as part of the Carbonfact Textile LCA Database. Its purpose is to provide fully traceable, transparent, and updatable LCI datasets for 1 kg of flat-knitted fabric, covering the relevant process variants in current commercial use. The aim is to replace generic dataset placeholders with LCIs that are scientifically robust, verifiable, and aligned with real-world textile production technologies.

The work involves a structured literature review of energy consumption, yarn losses, lubrication, and auxiliary systems in flat knitting, followed by the construction of parametric foreground inventories for three scenarios — best case (BCS), average, and worst case (WCS) — that bracket the industrial variability documented in the literature.

1.2 The flat knitting process

Flat knitting is a textile manufacturing operation in which two flat needle beds, arranged in a V-configuration, interlace yarn to produce shaped or rectangular fabric panels. In contrast to circular knitting, flat knitting machines operate with a reciprocating carriage that traverses the needle beds, allowing the formation of fully-fashioned garment panels, selvedge edges, and intarsia colour work without post-process cutting.

Industrial flat knitting machines range from coarse-gauge (GG 3–7) machines producing knitwear and outerwear, to fine-gauge (GG 14–18) machines for lightweight jersey and hosiery structures. Contemporary computerised flat knitting systems — such as those produced by STOLL, Shima Seiki, and Protti — integrate needle selection, yarn carrier control, and tension management in a single programmable platform, enabling near-zero cutting waste and panel-level customisation.

Despite the technological maturity of flat knitting, life-cycle inventory data for this specific technology remain scarce in the published literature. Most studies aggregate flat and circular knitting or report factory-level electricity averages without technology disaggregation. The EF 3.1 dataset library does not distinguish between flat and circular knitting at the process level. This document addresses that gap.

1.3 System boundary

The system boundary covers gate-to-gate flat knitting: from yarn input at the machine to finished knitted fabric panels at the mill gate. The following flows are **included** within the foreground model:

- Direct electricity consumption of the knitting machine (drive motors, carriage, needle selection electronics).
- Indirect energy for auxiliary systems: HVAC and humidity control, compressed air for pneumatic yarn guides and lint extraction, illumination, and automated needle-bed cleaning systems.
- Lubricant input and the corresponding waste mineral oil output.
- Capital goods: machine amortisation and associated end-of-life (EoL) flows.
- Building infrastructure: allocated floor area per kg of fabric.

The following are explicitly **excluded** from the foreground model:

- Fibre and yarn production (upstream).
- **Textile and material waste:** yarn losses and waste-yarn treatment are handled externally by the caller via the process-specific IOR. The IOR scales the upstream yarn input and routes the lost yarn to end-of-life treatment. No inline yarn-loss or textile-waste exchange is included in the dataset itself.
- Wet processing, finishing, dyeing, and garment assembly (downstream).
- Packaging and distribution.
- Building construction costs beyond the amortised area share.

1.4 Functional unit and scope

The functional unit is **1 kg of knitted fabric panel** at the flat knitting mill gate, ready for inspection and downstream processing. Background processes are linked via **ecoinvent 3.12, system model: cut-off by classification**, using generic market activities (GLO/RoW).

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2. Data Sources and Literature Review

2.1 Existing LCI datasets for knitting

The EF 3.1 dataset library does not contain a knitting-specific process dataset that distinguishes flat from circular technology. The closest available datasets — 'knitting service' entries in ecoinvent 3.12 — aggregate both technologies under a generic knitting activity, without documentation of the underlying measurement basis, machine type, or yarn fineness. Van der Velden et al. (2014) report an electricity consumption of approximately 1.16 kWh/kg for flat knitted fabrics derived from Dutch industry data, but without specification of gauge, yarn count, or machine configuration. Ecobalyse (ADEME & Ministère de la Transition Écologique, 2024) provides a parametric range of 1.2–2.4 kWh/kg for knitting, with 3.5 kWh/kg as the total plant electricity when auxiliaries are included, but does not distinguish flat from circular knitting.

This absence of technology-specific data is the primary motivation for constructing a dedicated flat knitting LCI based on a structured literature review.

2.2 Observations from the literature review

The following observations summarise the key findings from the reviewed literature and their implications for LCI modelling.

1. Flat-knitting-specific energy data are extremely scarce. Most studies either combine flat and circular knitting or report factory-level electricity averages without technology disaggregation. The most cited source, Van der Velden et al. (2014), provides a single point estimate at approximately 1.16 kWh/kg for flat knitted fabric; no study provides a continuous function of electricity versus yarn count for this technology.

2. The Ecobalyse range of 1.2–2.4 kWh/kg for knitting machine electricity is the broadest peer-reviewed parametric range available and is used here as the primary empirical envelope. Maity et al. (2021) report a compatible range of 1.2–2.1 kWh/kg for circular knitting machines, confirming that the lower bound is appropriate for efficient modern machines.

3. Yarn losses for flat knitting range from 1% to 5% in the reviewed literature (Laursen et al., 2007; Sandin et al., 2019; Rakib Ahmed, 2024). The 1.5% lower bound reflects modern computerised machines with minimal setup waste; the 5% upper bound reflects fine-gauge machines, complex shaping, and older equipment. Ecobalyse reports 3–5.5% for the combined flat/circular category. The average of 3% used in this model is conservative relative to the Ecobalyse range.

4. Lubricant consumption is consistently reported at 3–8 g/kg of fabric. Hasanbeigi & Price (2015) and Emdadul Haq et al. (2022) report 3–6 g/kg depending on machine gauge and lubricant system design. The Idemat (2012) database reports a default of 5 g/kg, which is used as the average-case value in this model.

5. Indirect energy — HVAC, compressed air, illumination, and cleaning — represents a substantial fraction of total plant electricity that is rarely sub-metered in published studies. Ecobalyse documents that machine electricity represents approximately 35% of total plant electricity in a knitting mill, implying that 65% is consumed by auxiliary systems. This ratio, combined with the 2.4 kWh/kg upper-bound machine electricity, yields an indirect energy estimate of 4.46 kWh/kg (derivation: 2.4 ÷ 0.35 × 0.65). This is treated as a shared default pending sub-metered primary data (see Section 3.3 and the shared methodology page).

6. Capital goods are rarely included in flat knitting LCIs. EF guidance (Fazio et al., 2020) requires capital goods and their EoL to be included unless excluded under the 3% mass cut-off with explicit justification. Weidema et al. (2013) describe the standard ecoinvent practice of amortising machine mass over lifetime output. This model applies that approach using OEM-verified machine mass data (see Section 3.4).

7. No published study provides LCIA results specifically for flat knitting that can be used as a direct external benchmark. Comparison is therefore made against the EF 3.1 generic knitting dataset and the Carbonfact circular knitting LCI.

3. Life Cycle Inventory Modelling

This section presents the foreground life cycle inventory for flat knitting. Three scenarios are defined — Best Case (BCS), Average, and Worst Case (WCS) — that bracket the industrial variability documented in the literature. All values are expressed per 1 kg of finished knitted fabric at the mill gate.

3.1 Inventory table

The table below presents the foreground inventory for all three scenarios. Dataset names refer to ecoinvent 3.12 (cut-off by classification). Yarn losses and textile-waste treatment are not included inline — they are handled by the caller via the IOR (see Section 3.5).

Flow	Average	BCS	WCS	Unit	Dataset	
Technosphere inputs						

Flow	Average	BCS	WCS	Unit	Dataset	
Lubricant, average	0.005	0.005	0.008	kg	`lubricating oil production {RoW} \	cut-off`
Waste mineral oil (lubricant EoL)	0.005	0.005	0.008	kg	`market for waste mineral oil {RoW} \	cut-off`
Electricity						
Machine electricity (direct)	1.20	1.16	2.40	kWh	`market group for electricity, medium voltage {GLO} \	cut-off` — Ecobalyse range 1.2–2.4 kWh/kg (ADEME 2024); BCS from Van der Velden et al. (2014)
Indirect energy (HVAC, compressed air, lighting, cleaning)	4.46	4.46	4.46	kWh	`market group for electricity, medium voltage {GLO} \	cut-off` — derived: 2.4 ÷ 0.35 × 0.65 = 4.46 kWh/kg. See Indirect Energy shared page.
Capital goods						
Building machine (capital good)	2.8×10^{-6}	8.3×10^{-6}	1.1×10^{-6}	unit	`market for building machine {GLO} \	cut-off` — 1000 kg ÷ Q_lifetime ÷ 2000 kg/unit. See Section 3.4.
Building infrastructure	1.39×10^{-4}	1.39×10^{-4}	1.39×10^{-4}	m^2	`market for building, hall {GLO} \	cut-off` — KARL MAYER STOLL CMS 503 ki L envelope with cell factor. See Section 3.4.
Waste flows (auxiliary only)						
EoL machine — steel fraction	0.0047	0.0167	0.0023	kg	`market for waste steel {RoW} \	cut-off` — 850 kg steel ÷ Q_lifetime. See Section 3.4.
EoL machine — other materials	8.3×10^{-4}	2.8×10^{-3}	3.4×10^{-4}	kg	`treatment of waste electric and electronic equipment, shredding {GLO} \	cut-off`

> **Note on yarn losses:** the foreground dataset does not include a yarn input or textile-waste exchange. Yarn losses are handled externally via the IOR (Section 3.5): the caller scales the upstream yarn input by the IOR and routes the lost fraction to `market for waste yarn and waste textile {RoW} | cut-off`.

3.2 Electricity model

Machine electricity consumption for flat knitting is modelled as a range rather than a continuous function of yarn fineness, because no peer-reviewed source provides a statistically robust dtex-to-kWh/kg relationship for flat knitting specifically. The three scenario values are:

- **BCS (1.16 kWh/kg):** derived from Van der Velden et al. (2014), representing an efficient modern plant.
- **Average (1.20 kWh/kg):** lower bound of the Ecobalyse parametric range, conservative relative to the documented spread.
- **WCS (2.40 kWh/kg):** upper bound of the Ecobalyse range, corresponding to fine-gauge machines and high-speed operation.

All three scenario values are anchored in the Ecobalyse documentation (ADEME & Ministère de la Transition Écologique, 2024), which represents the most comprehensive parametric review of knitting energy consumption currently available in the public domain. The BCS value is further supported by Van der Velden et al. (2014), who report measured electricity consumption for flat knitted fabrics at approximately 1.16 kWh/kg in a Dutch industrial context.

Future refinement should be based on sub-metered primary data from flat knitting facilities, with yarn count as the primary explanatory variable.

3.3 Indirect energy

> **Shared methodology:** Full derivation, evidence base, and subsystem distribution are documented in Indirect Energy in LCA — Shared Methodology (Knitting).

Auxiliary systems — HVAC and humidity control, compressed air for pneumatic yarn guides and lint extraction, illumination, and automated needle-bed cleaning — are estimated at **4.46 kWh/kg** across all three scenarios. This value is derived from the Ecobalyse documentation (ADEME & Ministère de la Transition Écologique, 2024), which states that machine electricity represents approximately 35% of total plant electricity in a knitting mill:

$$E_{\text{indirect}} = 2.4 \div 0.35 \times 0.65 = 4.46\ \text{kWh/kg}$$

This derivation is conservative — anchored to the WCS machine electricity level — and is treated as a shared default for all flat knitting scenarios pending sub-metered primary data.

3.4 Capital goods and building infrastructure

> **Shared methodology:** Capital Goods & End-of-Life (Machinery) in LCA — Full modelling framework, calculation logic, and default values. This section states only the flat-knitting-specific inputs.

> **Shared methodology:** Building Infrastructure in LCA — Full allocation framework for building area. The flat-knitting-specific value is 1.39×10^{-4} m²/kg (KARL MAYER STOLL CMS 503 ki L envelope with cell factor).

For flat knitting, the reference machine is a computerised flat knitting machine with a verified mass of approximately **1,000 kg**, based on published specifications for the Fengfan double-carriage computerised flat knitting machine (net weight ~1,000 kg; 2,500 × 850 × 1,900 mm), cross-referenced with Fibre2Fashion supplier data. Machine productivity for the average scenario is set at **2.5 kg/h**, consistent with industry benchmarks reported by Sino Finetex (2025) and Changhua Knitting Machine.

Parameter	BCS	Average	WCS
Machine mass (kg)	1,000	1,000	1,000
Service life (years)	10	15	20
Operating hours per day	8	16	24
Operating days per year	300	300	365
Productivity (kg fabric/h)	2.5	2.5	2.5
Lifetime output Q (kg)	**60,000**	**180,000**	**438,000**
Machine allocation (kg/kg fabric)	0.0167	0.0056	0.0023
ecoinvent units (÷ 2000 kg/unit)	$8.3×10^{-6}$	$2.8×10^{-6}$	$1.1×10^{-6}$

Machine EoL is modelled by splitting machine mass into a steel fraction (85% = 850 kg) and other materials (15% = 150 kg). For the average scenario:

- Steel EoL = 850 ÷ 180,000 = **0.0047 kg/kg fabric** → `market for waste steel {RoW} | cut-off`
- Other materials EoL = 150 ÷ 180,000 = **$8.3×10^{-4}$ kg/kg fabric** → `treatment of waste electric and electronic equipment, shredding {GLO} | cut-off`

Building infrastructure is allocated using the KARL MAYER STOLL CMS 503 ki L machine footprint (2,700 × 909 mm) plus a cell factor accounting for access aisles, yarn storage, and maintenance clearance, yielding **$1.39×10^{-4}$ m^2/kg fabric** → `market for building, hall {GLO} | cut-off`.

> **Source:** Machine dimensions confirmed from KARL MAYER STOLL CMS 503 ki L official brochure (stoll.com). Machine mass (1000 kg) based on verified OEM datasheets for comparable computerised flat knitting machines (Fengfan double-carriage, Fibre2Fashion listings). Productivity (2.5 kg/h) consistent with industry benchmarks (Sino Finetex 2025; Changhua Knitting Machine). Operational parameters (15 years, 16 h/day, 300 days/year) are mid-range engineering defaults consistent with Hasanbeigi (2010) and IPPC BREF (2003). Replace with primary factory data when available.

3.5 Yarn losses and Input–Output Ratio (IOR)

Yarn losses during flat knitting arise from thread breaks during the knitting cycle, edge waste at garment panel selvedges, machine setup trials, and yarn carrier transitions. Unlike weaving, flat knitting does not involve sizing, and yarn preparation losses are limited to winding and creel feeding.

Losses are modelled **externally** via the IOR, applied by the caller upstream of this dataset:

$$\mathrm{IOR} = \frac{1}{1-L}$$

where L is the yarn loss rate as a fraction. For every 1 kg of finished fabric, IOR × 1 kg of yarn must enter the knitting machine. The caller also routes the lost yarn fraction to the appropriate end-of-life treatment. **No yarn input or textile-waste exchange is included inline in the foreground dataset itself** — only auxiliary waste (lubricant EoL, machine EoL) is modelled inline.

Scenario	Loss (%)	IOR (kg/kg)	Input required (kg)	Loss per 1 kg output (kg)
BCS — Best Case	1.5%	1.0152	1.0152	0.0152
Average (default)	**3%**	**1.0309**	**1.0309**	**0.0309**
WCS — Worst Case	5%	1.0526	1.0526	0.0526

Sources: Laursen et al. (2007); Sandin et al. (2019); Rakib Ahmed (2024, NTNU thesis); Ecobalyse (ADEME & Ministère de la Transition Écologique, 2024): 3–5.5%.

> **How to apply:** for 1 kg of flat knitted fabric, multiply the upstream yarn input by the scenario-specific IOR. The difference (IOR − 1) kg is lost as waste yarn and is routed by the caller to `market for waste yarn and waste textile {RoW} | cut-off`. At average: 1 kg fabric requires 1.031 kg yarn input; 0.031 kg is lost. No inline yarn-loss or textile-waste exchange is included in this dataset.

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4. Life Cycle Impact Assessment

4.1 LCIA results by scenario

Climate change impacts (kg CO_2eq/kg fabric) were computed using Brightway2 with the IPCC 2021 GWP100 characterisation factors, applied to the foreground inventory defined in Section 3 with ecoinvent 3.12 (cut-off) background processes. The dominant contributor is indirect energy in all three scenarios, reflecting the high auxiliary electricity load estimated from the Ecobalyse derivation.

Dataset / Scenario	GWP (kg CO_2eq/kg)	Dominant contributor	Notes
Flat Knitting - Average	3.94	Indirect energy (~65%)	Subject to grid electricity mix; GLO average used
Flat Knitting - BCS	3.91	Indirect energy (~65%)	Lower machine electricity, same auxiliaries
Flat Knitting - WCS	4.77	Indirect energy + machine (~70%)	Higher machine electricity drives total up

Note: LCIA result ranges reflect sensitivity to electricity grid mix. For the GLO average electricity market, the contribution of the knitting process itself (excluding upstream yarn production) to total garment GWP is typically in the range of 5–15%, consistent with Van der Velden et al. (2014).

4.2 Comparison with EF 3.1

EF 3.1 does not include a knitting-specific dataset that distinguishes flat from circular technology. Comparison is therefore made against the generic knitting activity in ecoinvent 3.12 and against the Carbonfact circular knitting LCI.

The Carbonfact flat knitting LCI differs from generic knitting datasets primarily through the explicit modelling of indirect energy (4.46 kWh/kg), which is not separately accounted for in the EF 3.1 knitting entries. When indirect energy is excluded, the direct machine electricity values (1.16–2.4 kWh/kg) are broadly consistent with the ecoinvent dataset range. The inclusion of indirect energy increases total electricity consumption by approximately 65–85% relative to machine electricity alone, consistent with auxiliary load fractions reported for spinning mills (Hasanbeigi & Price, 2015) and weaving mills (Koç & Çinçik, 2010).

Capital goods contribute less than 1% of total GWP in all scenarios, consistent with the EF guidance expectation that machinery infrastructure is a minor contributor for energy-intensive processes.

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5. Inferences and Limitations

Flexibility of the Carbonfact flat knitting LCI model

The inventory is structured to be updatable at the parameter level — machine electricity, yarn loss rate, indirect energy ratio, and machine-specific capital goods can each be revised independently when primary data become available. The three-scenario structure provides a defensible range for sensitivity analysis and uncertainty communication in product-level LCAs.

The IOR-based approach to yarn losses ensures that upstream yarn impacts scale correctly with actual process efficiency, without requiring a fixed waste-yarn flow in the background dataset. This is consistent with the approach adopted across all knitting technology documents in this database.

Data gaps and priorities for improvement

Three data gaps represent the highest priority for future improvement. First, the absence of flat-knitting-specific electricity data as a function of yarn count (dtex) prevents dynamic scaling across yarn fineness classes. A dtex-to-kWh/kg relationship calibrated to flat knitting — analogous to the regression model developed for circular knitting in this database — would substantially improve the accuracy of product-level LCAs for garments with known yarn specifications.

Second, the indirect energy estimate (4.46 kWh/kg) is derived from a single-source ratio and applied uniformly across all scenarios. Sub-metered auxiliary energy data from flat knitting facilities — disaggregating HVAC, compressed air, lighting, and cleaning — would replace this with a more defensible, technology-specific value.

Third, the model does not currently distinguish between conventional computerised flat knitting and emerging whole-garment (WHOLEGARMENT/3D) flat knitting technologies. The 3D Knitting LCI in this database addresses the latter technology separately.

Limitations for external review

All energy values used in this model are from secondary literature sources published between 2007 and 2024. No primary mill data were collected for this version of the document. The indirect energy estimate carries the highest uncertainty (±30–50%) due to the single-source derivation and the absence of sub-metered data for flat knitting specifically.

Machine mass and productivity parameters are based on OEM datasheet values for specific machine models. These values are representative but may not reflect all machine configurations in commercial use. Capital goods contribute less than 1% of total GWP and are therefore not a material source of uncertainty for most applications.

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